

ex. 37, 38, (35-39)

& cosine Rule / Lashin's Theorem.

Pythagorean theorem

THEOREM NO. 2

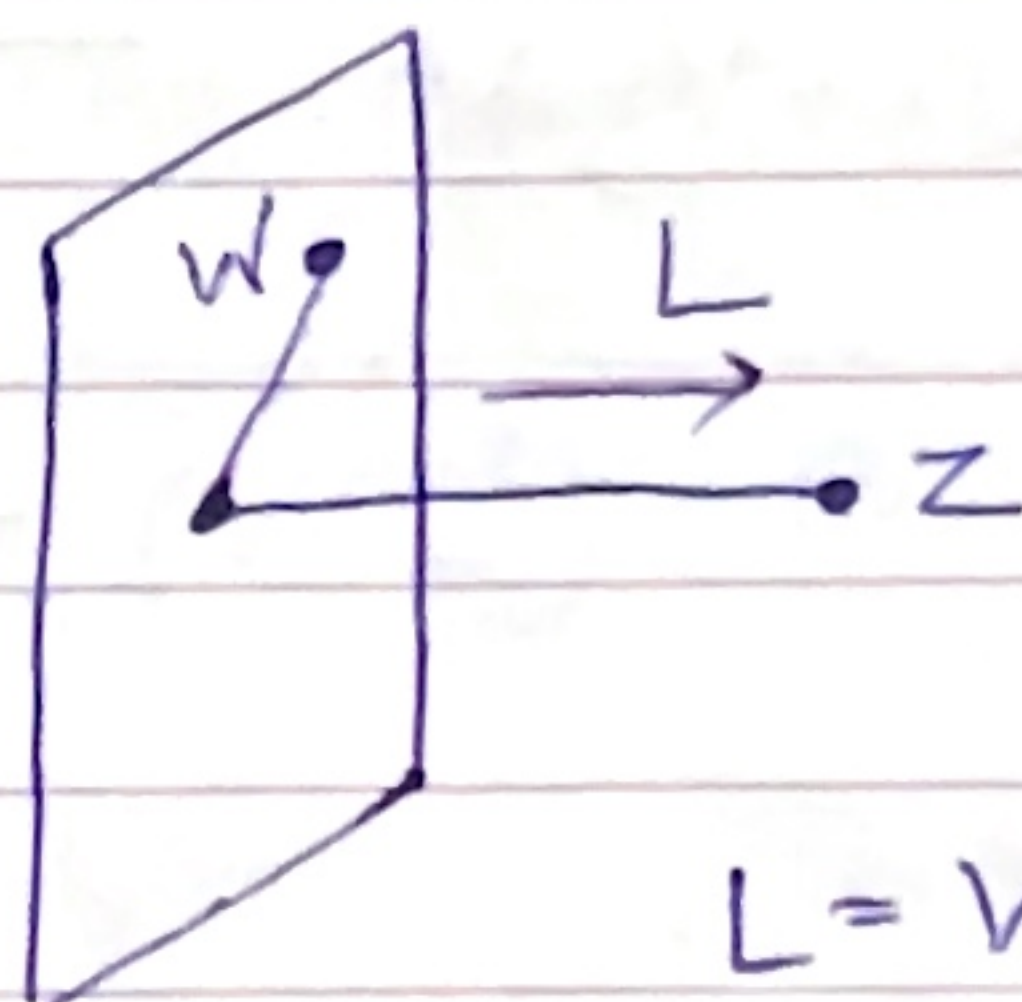
ORTHOGONAL COMPLEMENT

Note:

If a vector z is orthogonal to every vector in a subspace W , then z is said to be orthogonal to W .

* The set of all the vectors z that are orthogonal to W is called the orthogonal complement of W .

denoted as $W^\perp$



$$L = W^\perp \\ \text{and } W \cap W^\perp = \{0\}$$

Note

A vector x is in $W^\perp$ iff it is orthogonal to every vector in a set that spans W .

$W^\perp$ is a subspace of $\mathbb{R}^n$



KAGHAZ
www.kag haz.pk

"Theorem no 3"

let A be an $m \times n$ matrix.

The orthogonal complement of row space of A is null space and the orthogonal complement of column space is Null space of A^T

$\therefore$

$$(\text{Row } A)^\perp = \text{Nul } A$$

$$\& (\text{Col } A)^\perp = \text{Nul } A^T$$

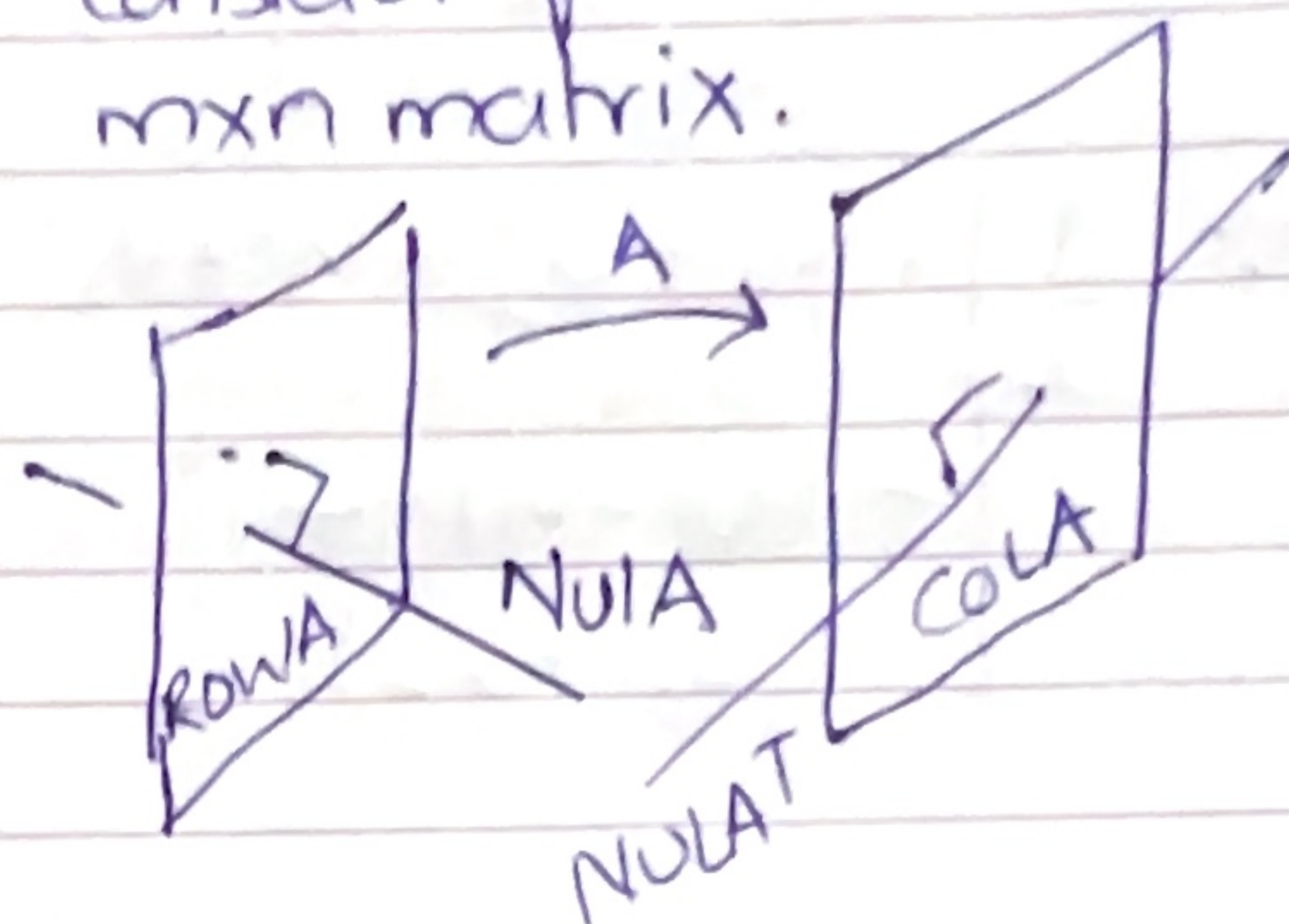
Remarks *

$\Rightarrow$ a common way to prove two sets are equal is to prove that one is the subset of another.

$\text{Nul } A = (\text{Row } A)^\perp$ is also established by showing that $\text{Nul } A$ is a subset of $(\text{Row } A)^\perp$ and $(\text{Row } A)^\perp$ is a subset of $\text{Nul } A$.

figure:

consider for a $m \times n$ matrix.

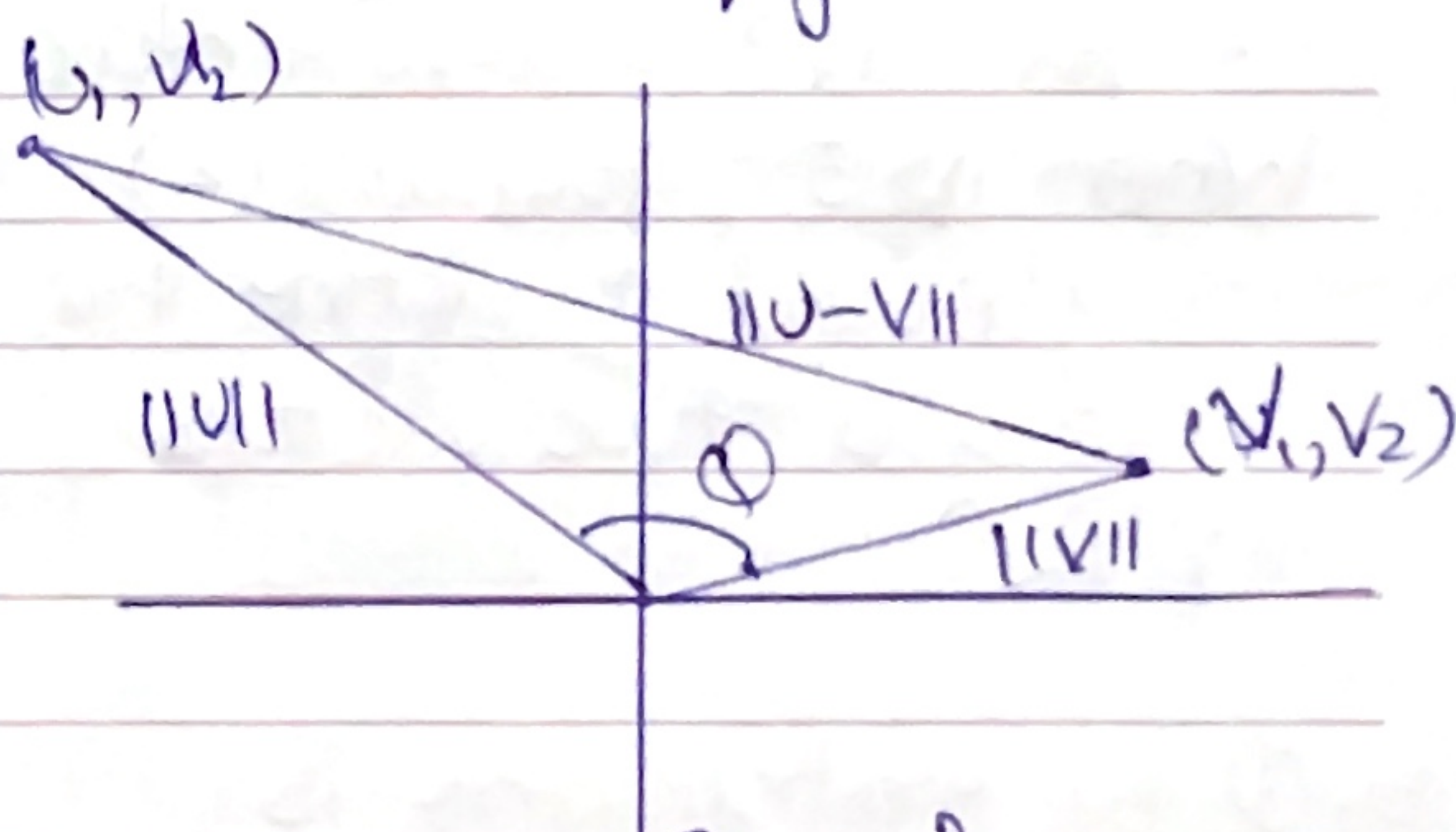


Angles in $\mathbb{R}^2$ and $\mathbb{R}^3$
(optional).

$$u \cdot v = \|u\| \|v\| \cos \phi. \quad \text{--- (1)}$$

where u/v are $\in \mathbb{R}^2 / \mathbb{R}^3$

to verify this formula at (1) consider the figure.



$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos \phi.$$

rearranging

$$\|u\| \|v\| \cos \theta \Rightarrow \frac{1}{2} [\|u\|^2 + \|v\|^2 - \|u-v\|^2]$$

open them
by magnitude
formula!

$$\Rightarrow \frac{1}{2} [u_1^2 + u_2^2 + v_1^2 + v_2^2 - (u_1 - v_1)^2 - (u_2 - v_2)^2]$$

EXERCISE 6.2ORTHOGONAL SETS.

$$\Rightarrow u_1 v_1 + u_2 v_2$$

$$\Rightarrow u \cdot v \quad \text{hence proved!}$$

— x —

Note:

$$u \cdot v = \|u\| \|v\| \cos \theta. \quad \text{--- (2)}$$

The verification of formula (2) for $\mathbb{R}^3$ is also similar. When $n > 3$, formula (2) may be used to define the angle b/w two vectors in $\mathbb{R}^n$.

$\cos \theta$ by statisticians is called correlation coeff.

A set of vectors $\{u_1, \dots, u_p\}$ in $\mathbb{R}^n$ is said to be orthogonal set if each pair of distinct vectors from the set is orthogonal i.e.

$$u_i \cdot u_j = 0 \quad \text{where } i \neq j.$$

use this to prove orthogonality of a set.

Theorem no. 4.

If $S = \{u_1, \dots, u_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$ then S is linearly independent and hence is a basis for the subspace by S .

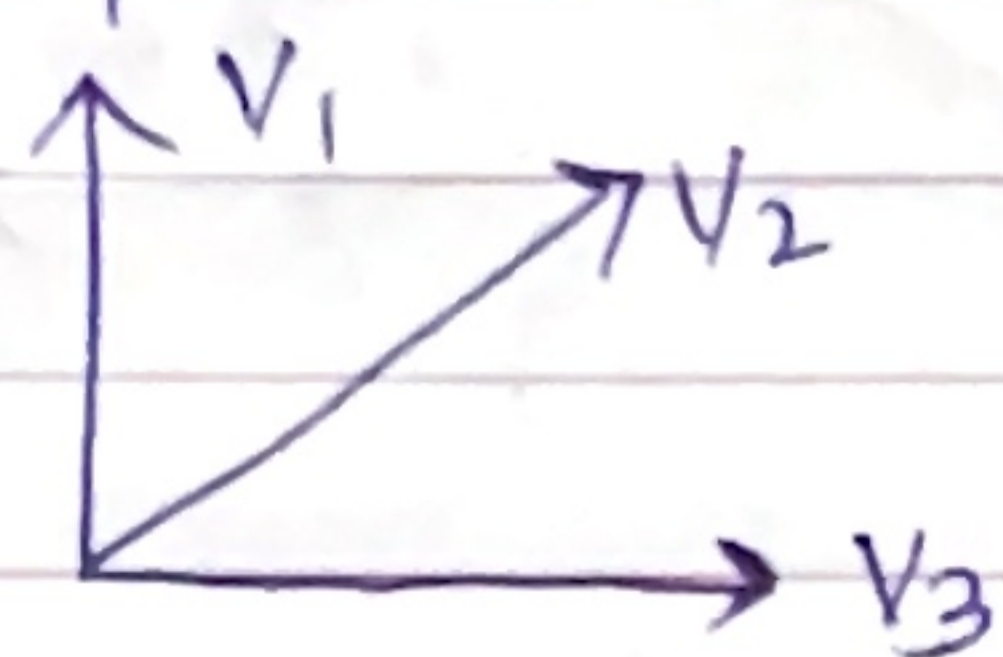
Therefore we also say that

Orthogonality always imply linear independence.

↓
even use $Ax=0$ to prove

→ however viceversa is not necessarily applicable.

Example



all three are L.I.D.N
But not orthogonal sets.

Orthogonal Basis:-

An orthogonal Basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Therefore the basis can be used to derive any vector of a subspace W whose orthogonal Basis were derived by.

$$y = c_1 b_1 + c_2 b_2 \dots c_p b_p. \quad (1)$$

* this basis is different from other basis because

$$y = B \bar{c} \quad x$$

$$y = [b_1 \ b_2 \ b_3 \dots] \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$$c_i = y b_i$$

Theorem no 5.

let b_i multiplies on b.s.

$$b_i y_i \Rightarrow b_i \cdot c_1 b_1 + b_2 \cdot c_2 b_2 + b_3 \cdot c_3 b_3 \dots b_n \cdot c_n b_n$$

$$b_i y_i \Rightarrow c_1 (b_i \cdot b_1) \dots 0$$

$$c_1 = \frac{b_i y_i}{b_i \cdot b_i}$$

∴ allowing formation of c's. more easily rather than explicit equation.

Question:

If V is an orthogonal Basis with $V = \{v_1, v_2, v_3\}$ express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as a

basis; as the linear combination of vectors U .

Note $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$

$$u_3 = \begin{bmatrix} -1/2 \\ 2 \\ -7/2 \end{bmatrix}$$

we need to show:

$$y = c_1 u_1 + c_2 u_2 + c_3 u_3$$

to do so, we need to find c_1 , c_2 , & c_3 .

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = 1, \quad c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = -2$$

$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = -2$$

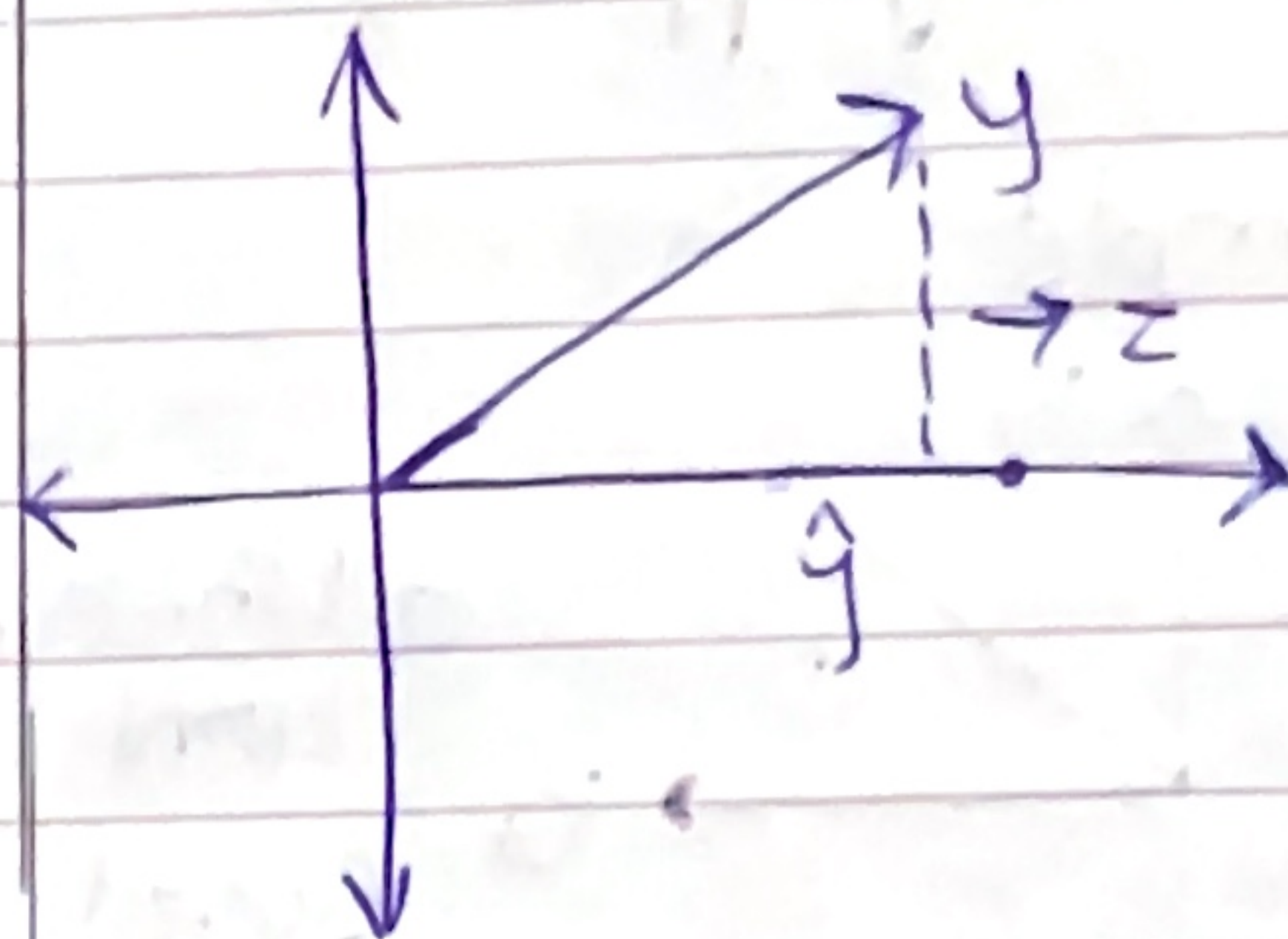
$$\therefore y = 1u_1 - 2u_2 - 2u_3$$

Ans!

6.2.

ORTHOGONAL PROJECTION

The decomposition of a vector into the sum of two orthogonal vectors is called as orthogonal projection.



We basically have the as stated above i.e. to break a vector into two parts.

(i) One part is in the direction of another vector u .

(ii) One part that is $\perp$ to the direction of that vector u .

This called as orthogonal projection

$$\text{i.e. } y = \hat{y} + z$$



Imagine that you have a vector v . you want to find the part of y that falls onto the / along the direction of v (like a shadow of y onto v).

$$(y - \alpha v) \cdot v = 0.$$

$$y \cdot v - \alpha v \cdot v = 0$$

$$y \cdot v = \alpha v \cdot v$$

$$\boxed{\alpha = \frac{y \cdot v}{v \cdot v}} \quad \text{---} *$$

The remaining part of y (what is left after moving the shadow) is called to be at the $\perp$ angle of v .

Now the formula for the projection is.

$$\hat{y} = \frac{y \cdot v}{v \cdot v} v$$

$\therefore \hat{y} \Rightarrow$ projection of y onto v .

$z \Rightarrow \perp$ to v .

$$\boxed{\text{Proj}_v y = \frac{y \cdot v}{v \cdot v} v} \quad \text{---} *$$

we can also say that.

$$\hat{y} \Rightarrow \alpha v \quad \text{where } \alpha \in \mathbb{R} \quad \text{---} \textcircled{1}$$

$\therefore \hat{y}$ is lying along v .

$$z = y - \hat{y} \quad \text{---} \textcircled{2} \text{ derived from original equation.}$$

$\rightarrow P \cdot T = 0$

If z & $\hat{y}$ are $\perp$ $\therefore$ dot product should be equal to 0.



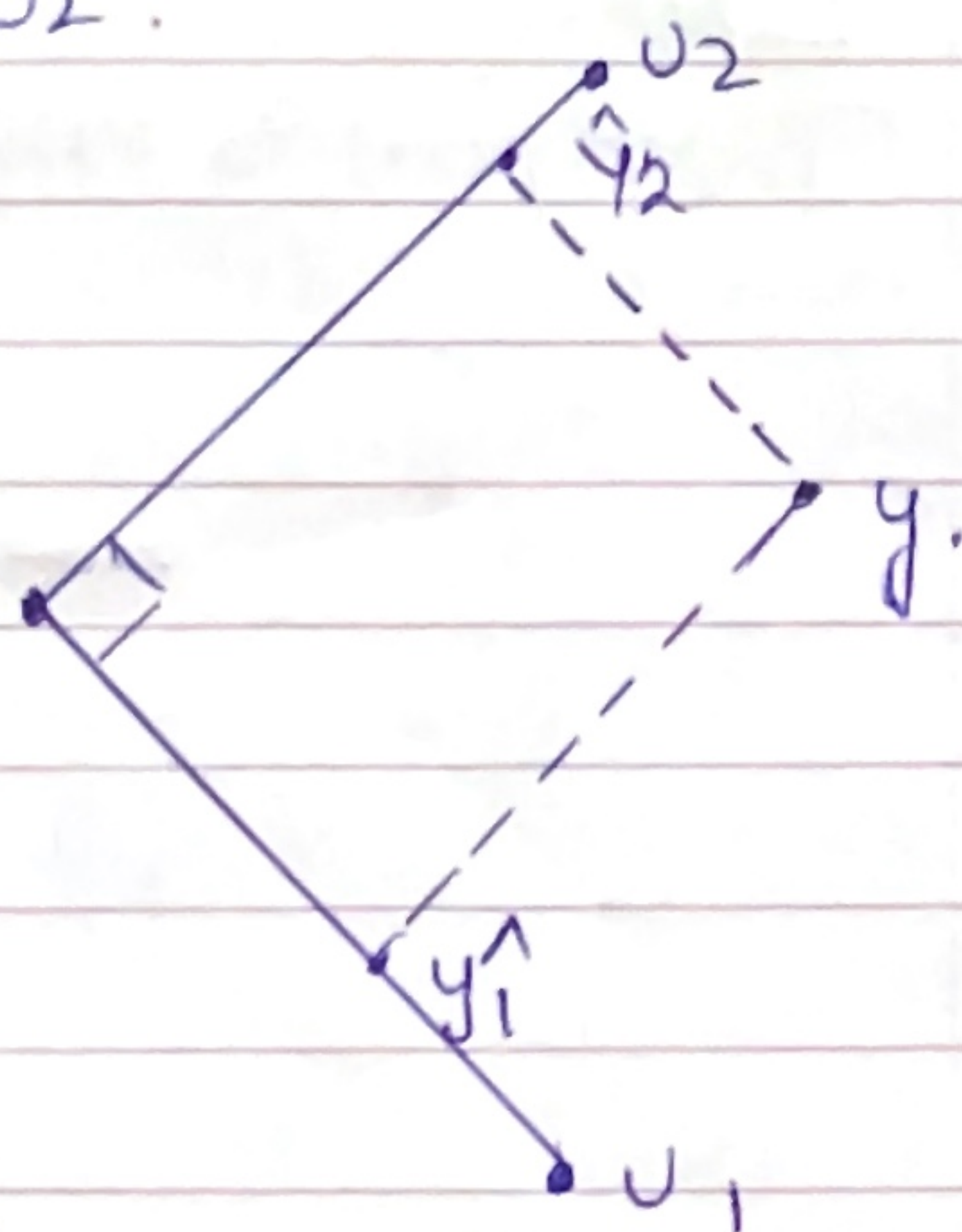
GEOMETRIC INTERPRETATION

Of Theorem no. 5

$$y = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2$$

The formula represents

- ① the projection of y onto the vector u_1 or subspace spanned by u_1 .
- ② the projection of y onto the vector u_2 or the subspace spanned by u_2 .



* Theorem no. 5

decomposes each y in span $\{u_1, \dots, u_p\}$ into the

sum of p projections onto 1 dimensional subspace that are mutually orthogonal.

ORTHONORMAL SETS.

A set $\{u_1, \dots, u_p\}$ is an orthonormal set if it is an orthogonal set of unit vectors.

If W is the subspace spanned by such a set, the $\{u_1, \dots, u_p\}$ is an orthogonal basis for W , since the set is automatically L.I.N.D. by theorem 4.

* The simplest example of an orthonormal set is the standard basis in $\mathbb{R}^n$ i.e. $\{e_1, \dots, e_n\}$.

* Any non empty set of $\{e_1, \dots, e_n\}$ is orthonormal too.

→ are proofs Theorems Included!

day / date:

→ learn procedure!

Example no 5.

→ to see if the
given set of vectors
is orthogonal/
orthonormal.

$\{v_1, v_2, v_3\}$

a. prove orthogonality.

$$v_1 \cdot v_2 = 0$$

$$v_2 \cdot v_3 = 0$$

$$v_3 \cdot v_1 = 0$$

} will be
proved!

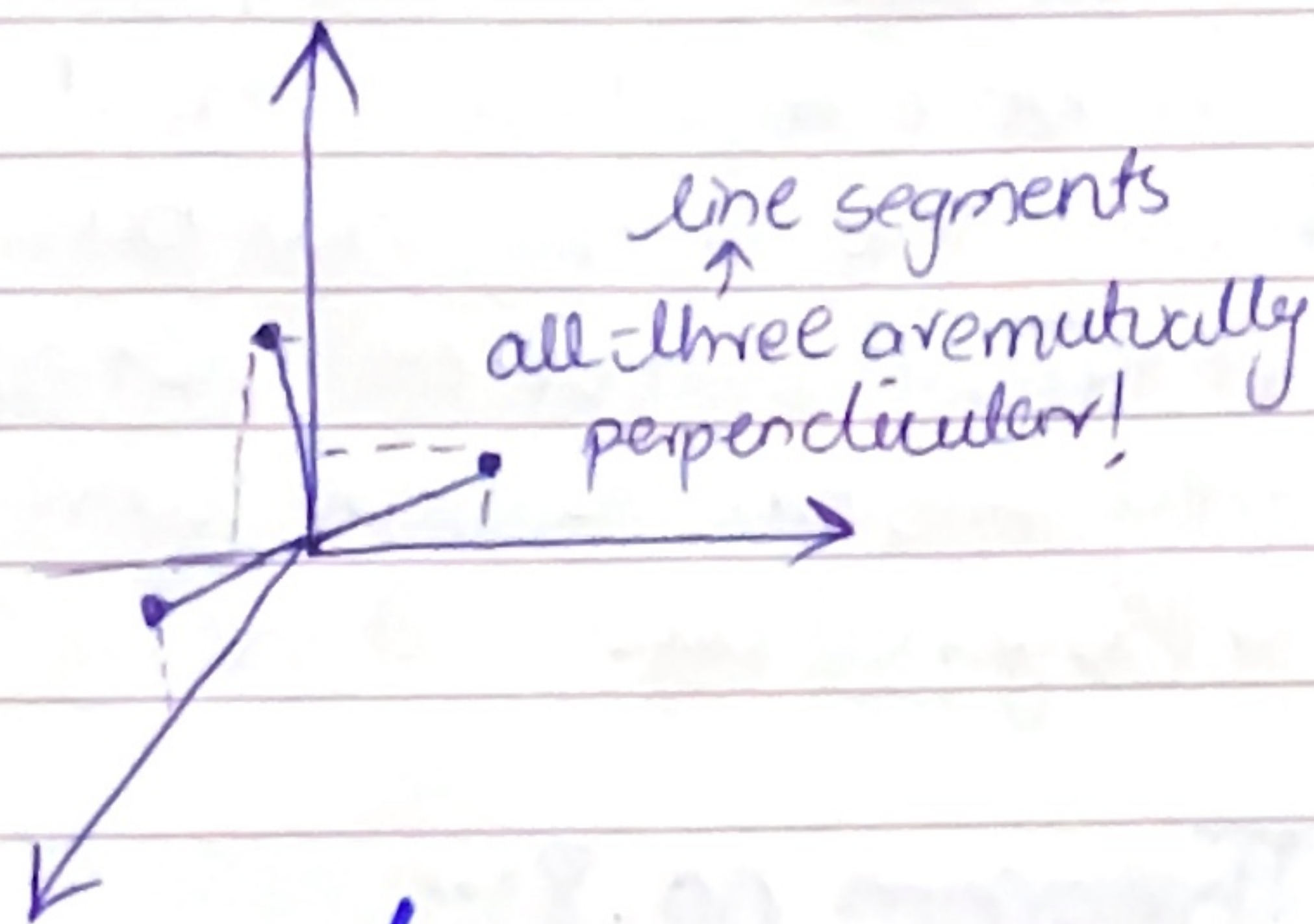
b. prove orthonormality.

$$v_1 \cdot v_1 = 1$$

$$v_2 \cdot v_2 = 1$$

$$v_3 \cdot v_3 = 1$$

} will be
proved!



6.
Theorem:-

An $m \times n$ matrix U has
orthonormal columns if
and only if $U^T U = I$.

Theorem 7:-

let U be an $m \times n$ matrix
with orthonormal columns
and let x and y be in $\mathbb{R}^n$

Then

$$(a) \|Ux\| = \|x\|$$

$$(b) (Ux) \cdot (Uy) = x \cdot y$$

$$(c) (Ux) \cdot (Uy) = 0$$

iff $x \cdot y = 0$.

* The property (a) & (c)
say that the linear
mapping $x \mapsto Ux$
preserves lengths and
orthogonality.

ORTHOGONAL MATRIX.

An orthogonal matrix
is a square matrix U

such that $U^{-1} = U^T$

* By theorem 6 such matrix
also has orthonormal cols.
* Such a matrix must have
orthonormal rows.

Example 7:-

To verify if the rows are orthonormal or not!

• Prove!

$$(i) \quad r_1 \cdot r_2 = 0$$

$$r_2 \cdot r_3 = 0$$

$$r_1 \cdot r_3 = 0$$

$$(ii) \quad r_1 \cdot r_1 = 0$$

$$r_2 \cdot r_2 = 0$$

$$r_3 \cdot r_3 = 0$$

~ x ~

EXERCISE 6.3

ORTHOGONAL PROJECTIONS

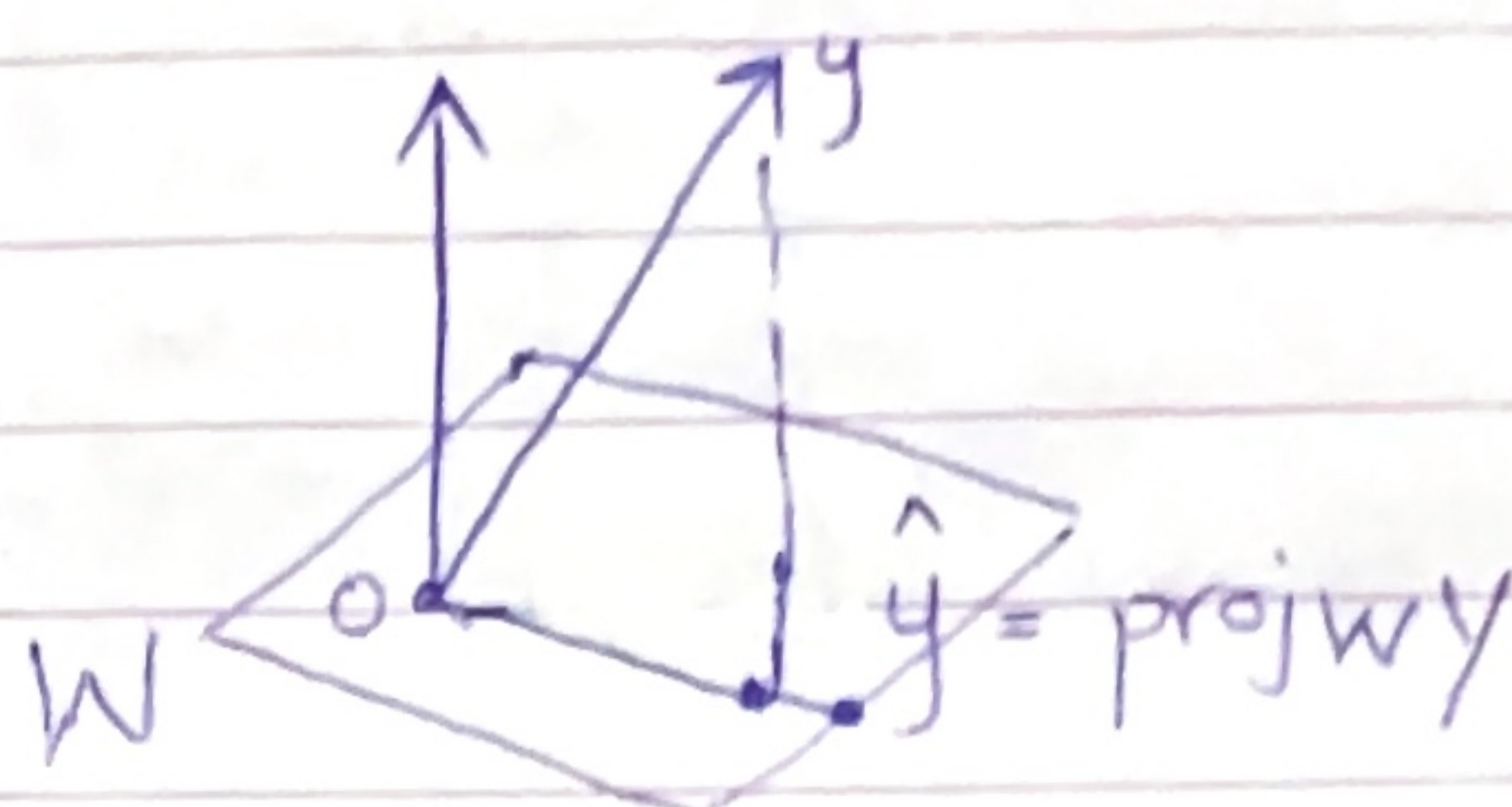
Consider orthogonal projection of a point in $\mathbb{R}^2$.

Given a vector 'y' and a subspace 'W' in $\mathbb{R}^n$, there is a vector $\hat{y}$ in W such that

(i) $\hat{y}$ is a unique vector in W for which $y - \hat{y}$ is orthogonal.

(ii) $\hat{y}$ is the closest ^{unique} vector in W (to y).

* These 2 properties of $\hat{y}$ provide the key to finding least squares solution of linear systems.



~ x ~

The next theorem shows that the decomposition $y = z_1 + z_2$ in Example 1 can be computed without having the orthogonal basis of $\mathbb{R}^n$, it can be computed only enough to have an orthogonal basis of W.

Theorem no. 8:-

Orthogonal Decomposition Theorem.

Let W be a subspace of $\mathbb{R}^n$. Then each y of $\mathbb{R}^n$ can be written uniquely as



$$y = \hat{y} + z. \quad \text{--- (1)}$$

where $\hat{y}$ is in W subspace and z is in $W^\perp$. In fact, if $(u_1 \dots u_p)$ is any orthogonal basis of W , then

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \dots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p \quad \text{--- (2)}$$

$$\frac{y \cdot u_p}{u_p \cdot u_p} u_p$$

$$\text{and } z = y - \hat{y}$$

(same figure as before).

~ x ~

* Similarly to proof that z is orthogonal to the subspace W .

let any orthogonal basis of W i.e. $\{u_1 \dots u_p\}$.

$$\text{then } (u_1/u_2 \dots /u_p) \cdot (z) = 0$$

stating that z is orthogonal to each vector in basis given.

~ x ~

To show uniqueness of decomposition in (1).

let 'y' can be written as.

$$y = \hat{y}_1 + z_1$$

with $\hat{y}_1$ in W and z_1 in $W^\perp$

also

$$y + z = \hat{y}_1 + z_1$$

The equality shows that

$$y - \hat{y}_1 = z_1 - z$$

$\downarrow \quad \quad \quad \downarrow$
 $\in W \quad \quad \quad \in W^\perp$

$\checkmark$ is in W and in $W^\perp$ (because z_1 and z are both $W^\perp$ and $W^\perp$ is a subspace) *

→ a vector $\in W$ & $W^\perp$ must be orthogonal to itself.

$\therefore v_1 \cdot v_1 = 0$ only orthogonal vector orthogonal to others and lies in that subspace is zero which shows that $v = 0$

This proves that $\hat{y} = \hat{y}_1$ and also $z_1 = z$ are unique.

The uniqueness of decomposition (2) shows that the orthogonal projection $\hat{y}$ depends only on W and not on the particular basis used in (2).

meaning:

The vector we get by projecting y onto W is always the same, no matter which basis you use for W .

"Only subspace matters - not how we describe it".

* you can change the basis of W but the projection will not change i.e. projections are invariant under the basis choice.

EXAMPLE NO. 2 :-

$$u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \text{ write } y \text{ as the sum of the/a vector in } W. \quad \text{--- (1)}$$

and a vector orthogonal to W . --- (2)

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} \cdot u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} \cdot u_2 \quad \text{--- (1)}$$

$$y - \hat{y} = \text{--- (2) sol.}$$

verification by

$$y = \hat{y} + z$$

$$y = \hat{y} + (y - \hat{y})$$

↳ put values derived here.

GEOMETRICAL INTERPRETATION OF ORTHOGONAL PROJECTION.

When W is 1-dim subspace the formula (2) for $\text{proj}_W y$ contains just one term.

↳ (A)

Thus, when $\dim W > 1$ each term in (2) is itself orthogonal projection of y onto a one-dim subspace spanned by one of the u 's in the basis for W .

↳ (B)

(A) states that we did:

$$\text{proj}_W y = \left(\frac{y \cdot u_1}{u_1 \cdot u_1} \right) u_1$$

if $W = \text{span} \{u_1\}$

⑤ states that more dim
'W' has more than 2 dim or
more than
with basis as:

$$W = \text{span} \{u_1 \dots u_k\}$$

then

$$\text{proj}_W y = \sum_{i=1}^k \left(\frac{y \cdot u_i}{u_i \cdot u_i} \right) u_i \dots$$

$$\left(\frac{y \cdot u_k}{u_k \cdot u_k} \right) u_k.$$

The orthogonal projections
of $\hat{y}$ of y onto W is the
sum of the projections
of y onto 1D subspaces
that are orthogonal to
each other.

~ x ~

Note:

If $\{u_1 \dots u_p\}$ is the
orthogonal basis of W and
if y happens to be in W ,
then the formula for

$\text{proj}_W y$ is exactly the
same as the representation
of y given as in theorem

(5)

in this case

$$\text{proj}_W y = y.$$

If y is in $W = \text{span} \{u_1 \dots u_p\}$
then $\text{proj}_W y = y.$

~ x ~

Theorem no 9

The Best Approximation Theorem

let W be a subspace of $\mathbb{R}^n$
let y be any vector in $\mathbb{R}^n$
and let $\hat{y}$ be the orthogonal
projection of y onto W . Then $\hat{y}$ is
the closest point in W to y
in that sense that

$$\|y - \hat{y}\| < \|y - v\| \text{ for all } v \text{ in } W \text{ distinct from } \hat{y}.$$

i.e $\|y - v\| \Rightarrow$ error and error
is minimized when.

$$v = \hat{y}$$

W

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2 \dots \frac{y \cdot u_p}{u_p \cdot u_p} u_p$$



THEOREM NO. 10.

If $\{u_1, \dots, u_p\}$ is a orthonormal basis for a subspace W of $\mathbb{R}^n$, then

$$\text{proj}_W y = (y \cdot u_1)u_1 + \dots + (y \cdot u_p)u_p$$

If $U = [u_1 \ u_2 \ \dots \ u_p]$
then

$$\text{proj}_W y = UU^T y \quad \text{--- (I)}$$

$$\forall y \text{ in } \mathbb{R}^n.$$

~ x ~

Suppose U is an $n \times p$ matrix with orthonormal columns and let W be the column space of U

then

$$U^T U x = I_p x = x \quad \forall x \in \mathbb{R}^p$$

↳ theorem 6.

$$UU^T y = \text{proj}_W y \Rightarrow \text{for all } y \text{ in } \mathbb{R}^n$$

↳ Theorem no. 10.

If U is a $n \times n$ square matrix with orthonormal columns, then U is an orthogonal matrix, the column space W is all of $\mathbb{R}^n$ and $UU^T y = Iy = y \quad \forall y \text{ in } \mathbb{R}^n$.

~ x ~

EXERCISE 6.4

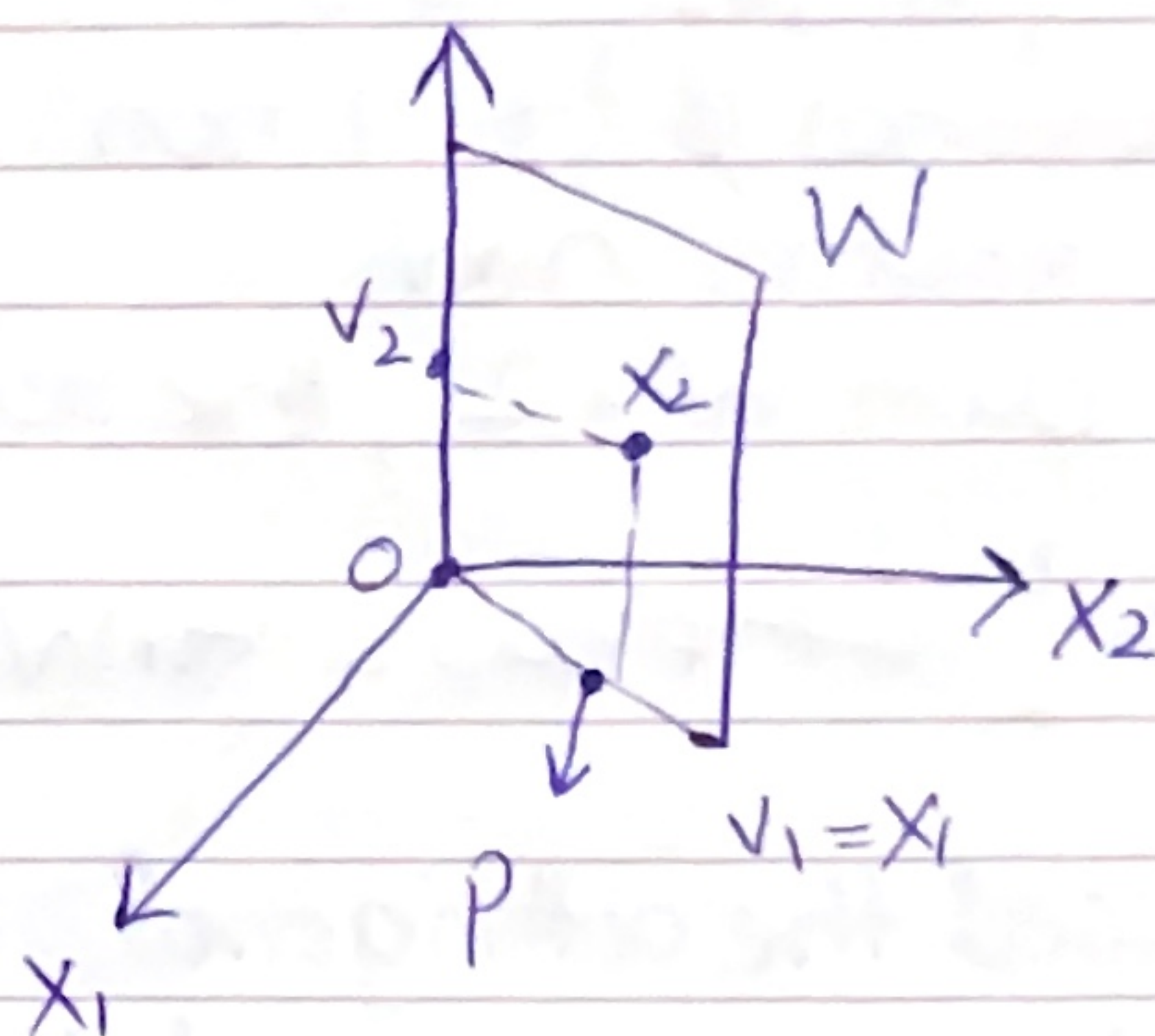
THE GRAM-SCHMIDT
PROCESS.

A simple algorithm for producing an orthogonal or orthonormal basis for any non-zero subspace of $\mathbb{R}^n$.

Example: - let $W = \text{span} \{x_1, x_2\}$

where $x_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ -

Construct an orthogonal Basis $\{v_1, v_2\}$ for W .



The subspace W is shown in the figure along with x_1 & x_2 and the projection of x_2 as p on x_1

Basically v_1 & v_2 are vectors of the subspace ^{day/date} ... our target to find them:-

The component of x_2 orthogonal to x_1 is

$x_2 - p$, which is in the subspace W (because it is formed from x_2 and a multiple of x_1)

let $v_1 = x_1$

then $v_2 = x_2 - p$

$$\Rightarrow x_2 - \frac{x_2 \cdot x_1}{x_1 \cdot x_1} x_1$$

$$v_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{15}{45} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

then $\{v_1, v_2\}$ is an orthogonal set of non zero vectors in W

Since $\dim W = 2$, the set $\{v_1, v_2\}$ is the basis for W .

$\Rightarrow$ we used the orthogonal projections concept to derive basis for our subspace.

EXAMPLE NO. 2

$$\text{let } x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Then $\{x_1, x_2, x_3\}$ is a l.i.n.d and thus the basis for a $\mathbb{R}^3$ space.

Construct an orthogonal Basis for W .

Step no. 1.

let $v_1 = x_1$ &

$$W_1 = \text{span}\{x_1\} = \text{span}\{v_1\}$$

Step no. 2

let v_2 be a vector produced by subtracting x_2 from the projection of x_2 on subspace W_1

$$\text{i.e. } v_2 = x_2 - \text{proj}_{W_1} x_2$$

$$\Rightarrow x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$



$$V_2 \Rightarrow \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

Step no. 3: appropriately scale V_2 (optional).

$$V_2 \Rightarrow \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Step no. 3

let V_3 be a vector produced by subtracting X_3 from its projection onto the subspace W_2 .

Use the orthogonal basis

$\{V_1, V_2\}$ to compute onto the W_2 .

$$\text{proj}_{W_2} X_3 = \frac{X_3 \cdot V_1}{V_1 \cdot V_1} V_1 +$$

$$\frac{X_3 \cdot V_2}{V_2 \cdot V_2} V_2.$$

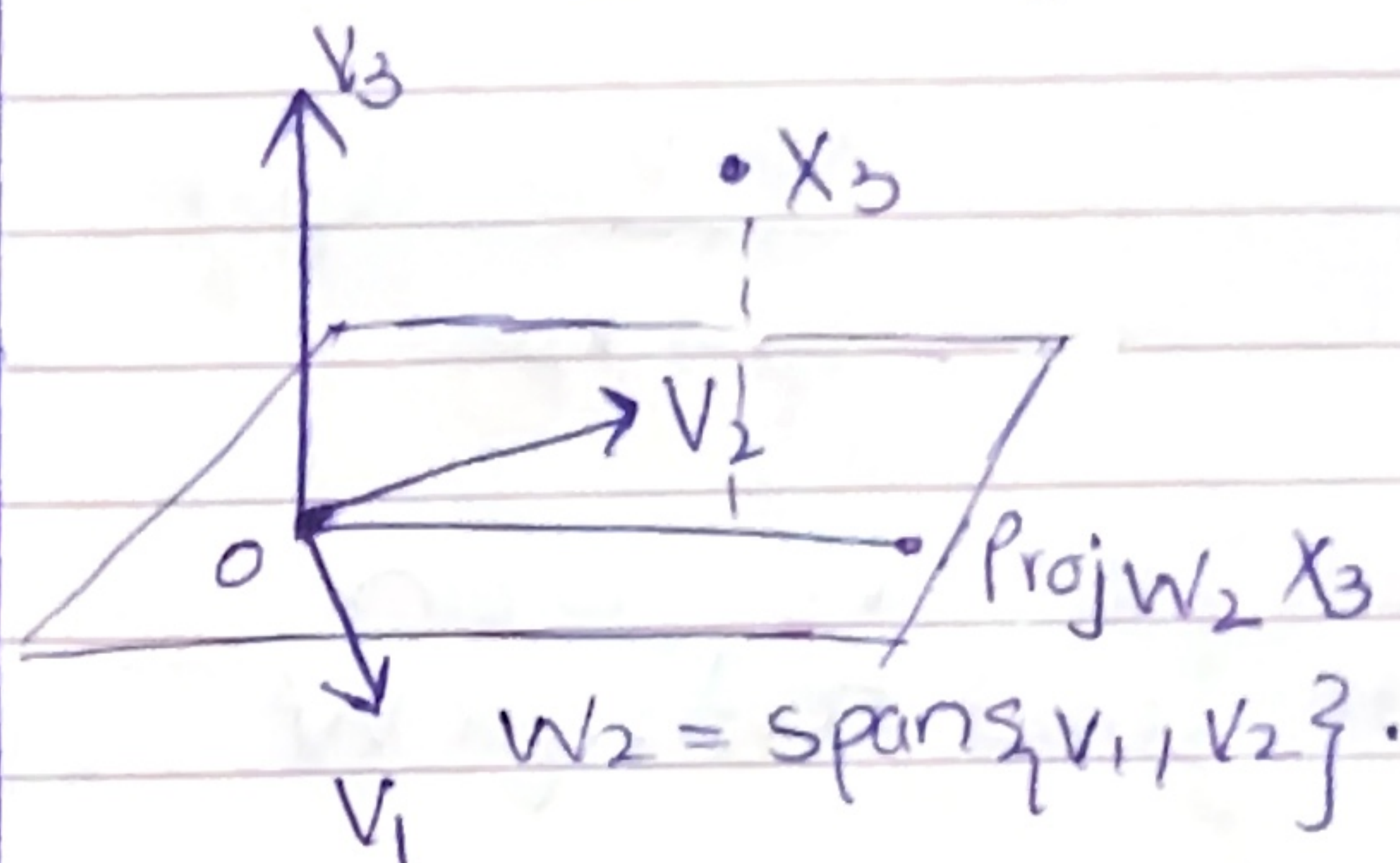
$$\Rightarrow \begin{bmatrix} 0 \\ 2/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

and finally

$$V_3 = X_3 - \text{proj}_{W_2} X_3$$

$$\Rightarrow \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

Now analyze the figure.



Observe that V_3 is in W because X_3 and $\text{proj}_{W_2} X_3$ are both in W .

$\therefore \{V_1, V_2, V_3\}$ is an orthogonal set of non zero vectors and hence a L.I.N.D set in W .
 $\therefore$ a basis for W .

THEOREM NO. 11.

The Gram-Schmidt Process.

Given a basis $\{x_1, \dots, x_n\}$ for a non zero subspace W in $\mathbb{R}^n$

★ Note Orthogonal Basis for row & col space are completely different. day / date:

define

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$v_p = x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_p \cdot v_2}{v_2 \cdot v_2} v_2 \dots$$

$$\frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} v_{p-1}$$

Then $\{v_1 \dots v_p\}$ is an orthogonal basis for W

In addition $\text{span}\{v_1 \dots v_k\} = \text{span}\{x_1 \dots x_p\}$
for $1 \leq k \leq p$

ORTHONORMAL BASES:

An orthonormal bases is constructed from a orthogonal bases. Simply normalize it (i.e scale it) with all v_k .

Example

$$v_1 = \begin{bmatrix} 3 \\ 8 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

the orthonormal basis is given by.

$$u_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{45}}$$

$$u_2 = \frac{1}{\|v_2\|} v_2$$

QR FACTORIZATION OF MATRICES:-

If a $m \times n$ matrix has linearly independent cols $x_1 \dots x_n$ then applying the Gram-Schmidt process (with normalization) to $x_1 \dots x_n$ amount to factoring A .

"If A is an $m \times n$ matrix with L.I.N.D cols, then A can be factored as $QR = A$ where Q is an $m \times n$ matrix whose cols form a orthonormal basis of $\text{Col } A$ and R is an $n \times n$ upper triangular matrix (invertible)

with positive entries
on its diagonal.

Example no. 4:-

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

find QR factorization

⇒ the cols of A are the
vectors x_1, x_2, x_3

An orthogonal Basis for $\text{col}(A)$

⇒ $\text{span} \{x_1, x_2, x_3\}$

as previously found by e.g no. 2

⇒ Imp!

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 1 \end{bmatrix}$$

Scaling $v_3 \Rightarrow$

$$\begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

Now normalizing given

v_1, v_2, v_3 as per example
method i.e

$$u_1 = \frac{1}{\|v_1\|} v_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

and so on.

$$\therefore Q = \begin{bmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{bmatrix}$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $u_1 \quad \quad u_2 \quad \quad u_2$

converted as orthonormals
now.

To find R.

we know that if Q is ortho-
normal then

$$Q Q^T = I.$$

∴ $A = QR$
multiplying Q^T on b.s

$$Q^T A = Q Q^T R.$$

$$Q^T A = IR$$

$$Q^T A \Rightarrow R$$

$$\begin{bmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R \Rightarrow \begin{bmatrix} 2 & 3/2 & 1 \\ 0 & 3/\sqrt{12} & 2/\sqrt{12} \\ 0 & 0 & 2/\sqrt{6} \end{bmatrix}$$

Ans!

Chapter no. 6

"LEAST SQUARES
PROBLEM"

In an inconsistent system when no solution is possible, the best one can do is to find an x that make Ax as close to b as possible.

like approximating Ax to b . The smaller the distance between b and Ax given by $\|b - Ax\|$ the better approximation.

This finding of x is known as the general least squares problem.

↪ arises from the fact that $\|b - Ax\|$ is square root of sum of squares.

if A is $m \times n$ and b is in $\mathbb{R}^m$, a least square solution of $Ax = b$ is an $\hat{x}$ in $\mathbb{R}^n$ such that

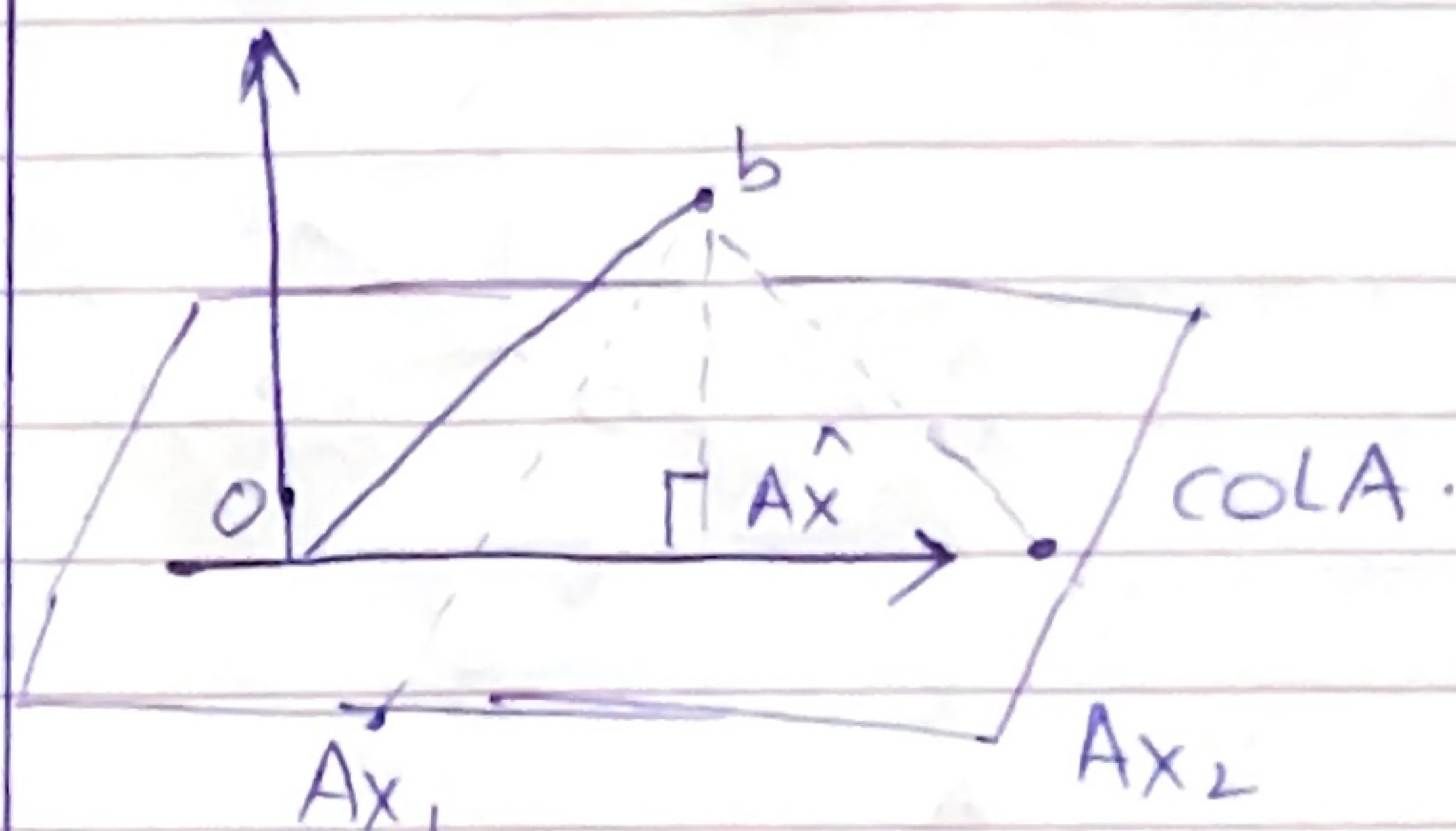
$$\|b - A\hat{x}\| \leq \|b - Ax\|$$

for all x in $\mathbb{R}^n$.

Note:

No matter what x we select the vector Ax will necessarily be in the column space, $\text{Col } A$

∴ we seek x that makes Ax the closest point in $\text{Col } A$ to b .



Solution of Least Squares Problem.

$$\text{Say } \hat{b} = \text{proj}_{\text{Col } A} b$$

because $\hat{b}$ is in the $\text{Col space of } A$, the equ. $Ax = \hat{b}$ is consistent.

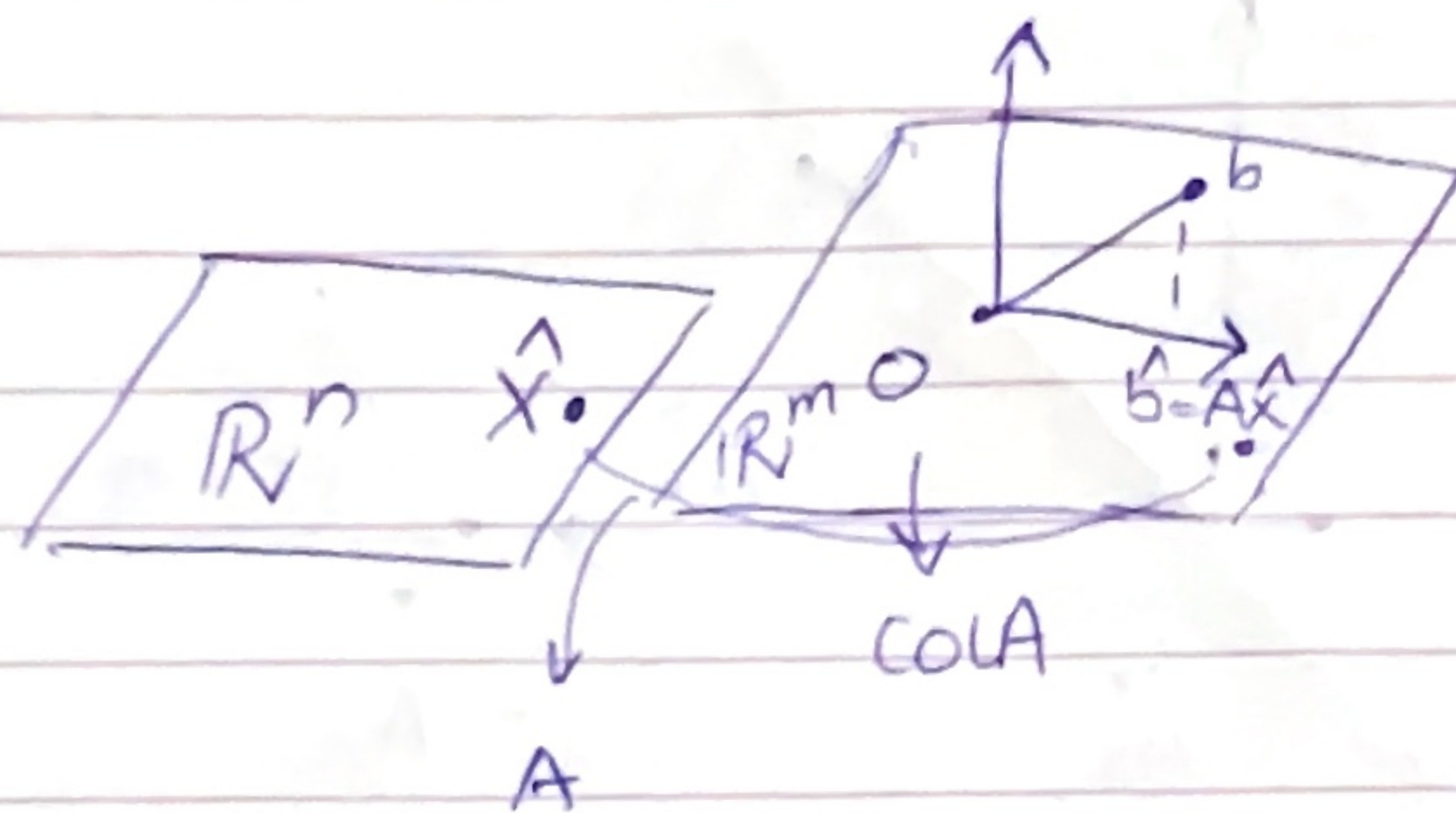
and there is an $\hat{x}$ in $\mathbb{R}^n$ such that

$$A\hat{x} = \hat{b} \quad (1)$$

Since $\hat{b}$ is the closest point in $\text{col } A$ to b , a vector $\hat{x}$ is a least square solution of $Ax=b$ if and only if $\hat{x}$ satisfies.

①

* Such $\hat{x}$ in $\mathbb{R}^n$ is a list of weights that will build $\hat{b}$ out of the cols of A



① can be orthogonally decomposed as

$b - \hat{b}$ is orthogonal to $\text{Col } A$, so $b - A\hat{x}$ is orthogonal to each column of A .

If a_j is any point in $\text{Col } A$

then

$$a_j^T (b - A\hat{x}) = 0.$$

$$\text{and } a_j^T (b - A\hat{x}) = 0$$

Since a_j^T is a row of A^T

$$A^T (b - A\hat{x}) = 0 \quad \text{--- ②}$$

$$A^T b - A^T A \hat{x} = 0$$

$$A^T A \hat{x} = A^T b. \quad \text{--- ③}$$

↳ This equ. represents a group / system of equations called: normal equations of $Ax=b$.

The sol. of ③ is often denoted as $\hat{x}$.

THEOREM 13:-

The set of least squares solutions of $Ax=b$ coincides with the nonempty set of solution of normal equations $A^T A x = A^T b$

EXAMPLE NO. 1.

Find a least square solution of an inconsistent system $Ax=b$.

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

Use normalize eqns)

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$A^T A x = A^T b.$$

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\Rightarrow \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$\Rightarrow \frac{1}{84} \begin{bmatrix} 84 \\ 168 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Ans!

Example no. 2

Similarly solves for a solution using augmented matrix.

THEOREM NO. 14 :-

Let A be a $m \times n$ matrix. The following statements are logically equivalent.

- The eqn. $Ax=b$ has a unique least square solution for each b in $\mathbb{R}^m$.

Exercise 27-29 imp!

day / date:

b. The cols of A are linearly independent.

c. The matrix $A^T A$ is invertible.

When these statements are true, the least square solution of $\hat{x}$ is given by

$$\hat{x} = (A^T A)^{-1} A^T b.$$

When the least squares sol. $\hat{x}$ is used to produce $A\hat{x}$ as an approximation to b , the dist from b to $A\hat{x}$ is called least squares error.

Example: Give A & b from example 1, determine the least squares error in the least squares sol. of $Ax = b$.

$$A\hat{x} = \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix}$$

$$\text{Now } \|b - A\hat{x}\| \Rightarrow \sqrt{84}.$$

ALTERNATIVES CALCULATIONS OF LEAST SQUARE SOLUTIONS.

EXAMPLE 4:-

find a least square solution of $Ax = b$.

for

$$A = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}, b = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$$

$$\hat{b} = \frac{b \cdot a_1}{a_1 \cdot a_1} a_1 + \frac{b \cdot a_2}{a_2 \cdot a_2} a_2$$

$$\Rightarrow \frac{8}{4} a_1 + \frac{45}{90} a_2$$

$$\Rightarrow \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \\ 1/2 \\ 7/2 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 \\ 1 \\ 5/2 \\ 9/2 \end{bmatrix}$$

Now that $\hat{b}$ is known we can solve $\hat{b} = A\hat{x}$. But this is trivial since we already know the weights to the cols of A to produce $\hat{b}$.

$$\hat{x} = \begin{bmatrix} 8/4 \\ 15/90 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

THEOREM 15:-

Given an $m \times n$ matrix with linearly independent cols let $A = QR$ be a QR factorization as in theorem 12.

Then for each b in $\mathbb{R}^m$, the eqn $Ax = b$ has a unique least square sol. given by $\hat{x} = R^{-1}Q^T b$.

EXERCISE 6.6

MACHINE LEARNING AND LINEAR MODELS

Machine learning:-

user linear models in situations where the machine is being trained to predict the outcome (dependant variables) based on the values of the inputs (independant variables).

One type of learning is to fit a curve such as the least squares lines or parabola, to the data.

Once the machine has learnt the pattern from the training data, it can estimate the value of the outputs based on given value of the input.

Least Square lines

let $Ax = b$ as

$$XB = y$$

$X \Rightarrow$ design matrix

$\beta \Rightarrow$ parameter vector

$y \Rightarrow$ observation vector.

The simplest relation b/w two variables x and y is through the equ. (linear).

$$y = \beta_0 + \beta_1 x$$

experimental data often produce $(x_1, y_1) \dots (x_n, y_n)$ that graphed to show close to a line.

β_0 and β_1 determine to make them as close to the points as possible.

let y_j - observed value of y .

ξ

$\beta_0 + \beta_1 x_j$ the predicted value of y .

The difference b/w predicted and observed value is called as residual.

There are many methods to measure how close the line is to the data;

1. is that adding the squares of the residuals.

The least square, is the line that minimizes the sum of the squares of residual. $\downarrow$ $(y = \beta_0 + \beta_1 x)$ also called as the 'line of regression of y on x '

$\Rightarrow$ any error in the data is assumed to be along the y -coordinates.

$\beta_0, \beta_1 \Rightarrow$ regression coefficients